Nolifa
Project breakdown

Three apps for South African farms, households and drivers, built on one shared engine. Made for the FNB App of the Year theme: climate resilience and essential services.

Nolifa Grow | Nolifa Drop | Nolifa Delivery

1. The idea in one page

One proposition: community resilience. When disease, drought or an outage hits, a community loses food and water first. Nolifa keeps three things working together: **Protect** (farm health and outbreak warning), **Prepare** (water forecast and asking for help together) and **Provide** (local farms keep food moving, with escrow and a delivery PIN). Everything else exists to make one of those three trustworthy.

What makes it different

- **Health first.** The farmer's app opens on the leaf scanner and the cattle symptom checker, not on a shop.
- **Trust by design.** Escrow, delivery PIN, disputes, verified farms, inspected land, and proof-backed community posts.
- **Honest about offline.** The scanner, symptom checker, herd register and crop tracker work with no signal. Orders, water readings, the community and maps need a connection. Deliveries, trucks and inspections need another person. The app has a page for this.
- **Built for rural and urban.** Inspection is tiered so a remote smallholder can start earning without waiting for a surveyor.

The three apps

App	For	Tabs	Folder
Nolifa Grow	Farmers	Home (scan crop, check animal, crop health AI, water radar), Orders, Alerts, Settings	~/nolifa-ios
Nolifa Drop	Customers	Shop, Water, Alerts, plus a menu (orders, wallet, settings, trust, help)	~/ubuntu-drop
Nolifa Delivery	Drivers	Jobs, Earnings, Account	~/nolifa-delivery
ubuntu_core (shared)	All three	Fees, escrow, PINs, disputes, water radar, theme, farms, shop and business screens	~/nolifa-ios/packages/ubuntu_core

2. How an order moves

Step	Who	What happens
1	Farmer	Registers, gets verified, pays a plan, shows land with photos, lists stock
2	Customer	Browses farm cards in Nolifa Drop, fills a one-farm basket, pays into escrow
3	Farmer	Gets a notification, packs the order, chooses to deliver it or request a driver
4	Driver (or farm)	Navigates to the farm, takes a pickup photo, navigates to the customer, takes a delivery photo
5	Customer	Gives the 4-digit PIN at the door. Delivery completes
6	Money	Customer confirms (or 2 hours pass) and escrow pays the farm and driver. A problem freezes the money and opens a dispute

3. Nolifa Grow (the farmer)

Home

- **Recent scans** at the top, kept small.

- **Scan a crop:** camera, on-phone diagnosis (online first, offline model as fallback), saved to history and added to the outbreak map by region.
- **Crop health AI:** add plots, track growth stage and days to harvest, set up sprinkler plans (see section 8).
- **Check a sick animal:** tick symptoms, get ranked conditions with controlled diseases flagged first. A **Your herd** button opens the herd register.
- **Water radar** (large) and a **Water provider** card for farms that can help communities.

Orders

- Earnings, orders to send out, on the way, and pending money.
- Notifications built from orders, orders received (who, what, what you get), stock management.
- Selling status with limits and inspection progress. Self-delivery actions: out for delivery, enter the buyer's PIN.

Alerts

- **Alarm** top left, pinging red when anything needs attention. It opens a page with water first and biggest, then orders, urgent outbreaks and farm alerts.
- **Hexagon outbreak map** with heat colours and Crops/Livestock filters. Tap any hexagon to see what is affecting it and how it is being handled.
- **Report an outbreak**, then the **Community**: posts must carry proof (AI leaf scan, animal symptom check plus photo, or water photo). Filters: Crops, Livestock, Both, Water. The water filter shows municipality updates only when a provider has answered.
- Outbreaks in the farmer's area, with details and advice.

Settings

- Personal and farm details, region, farm type, delivery fee, notifications, language (English and isiZulu), appearance, Business page, wallet, trust and safety, sign out.

4. Nolifa Drop (the customer)

- **Shop** is laid out like a food app: delivery address, cart, search, category chips, a promo carousel (Fresh near you, Rescue deals, Ready for outages), a water alert, then **farm cards** with the farm name in bold, rating, delivery method and offerings with quick add.
- Farms show photo banners and emblem logos made with Higgsfield. A basket holds one farm at a time and asks before starting a new one.
- **Water:** community reservoir level, 7-day forecast, alerts, and the community vote that can send a request to the provider.
- **Alerts:** water, supply and outbreak alerts that can fill the cart with a prepared bundle.
- **Menu:** orders, wallet and receipts, settings (address, phone, notifications), trust and safety, help, account and mode, sign out.
- On opening: a Welcome back notification, and a pop-up with a Vote button if the home community's water is low.

  

Examples of the farm banners used on the Shop home (generated with Higgsfield).

5. Nolifa Delivery (the driver)

- Driver sign-up, welcome cards, and verification (ID and vehicle checks) before going online.
- **Jobs:** open jobs nearby. After accepting: **Navigate** opens Apple Maps or Google Maps to the farm, then to the customer.
- Pickup and drop-off both **open the camera** for a proof photo. Delivery completes only when the customer's PIN matches.
- **Earnings:** total, recent deliveries, payouts, and how pay works. Drivers keep 80% of the delivery fee.
- SOS safety line, wallet, account and sign out. No welcome notification, by design.

6. Money rules

Item	Rule	Paid by
Buyer service fee	3% of items, minimum R2, maximum R15	Customer
Delivery fee	R25 base (Nolifa Delivery). A farm that delivers itself sets its own	Customer
Driver share	80% of the delivery fee. Nolifa keeps 20%	From delivery fee
Seller commission	8% Starter, 5% Grow, 3% Co-op, only on completed sales	Farm
Farm plan (platform fee, includes support)	Starter R49, Grow R99, Co-op R199 per month. Required to sell	Farm
Break-evens	Grow beats Starter above about R1 700 of monthly sales. Co-op beats Grow above about R5 000	

Worked example: a R129.03 order splits R92.92 to the farm, R20.00 to the driver and R16.11 to Nolifa (about 12.5% of the basket).

7. How a farm starts selling

A surveyor-first rule would shut out remote smallholders: long waits, travel costs, a lost season. So inspection is required but tiered. Full reasoning is in docs/INSPECTION.md.

Step	What happens
1. Register	Individual, co-op or company
2. Verify	ID, payout account, farm location
3. Plan	Monthly platform fee with support
4. Show your land	Geo-tagged photos of the field, crop and water source. Works offline
5. Sell with limits	Opens immediately: fresh whole produce only, up to R2 000 of sales, 45 days

Step	What happens
6. Inspection	Surveyor visit (about 14 days), extension officer (about 7), or co-op inspection day (about 21). Passing lifts the limits and allows eggs, milk and processed goods. A failure pauses listings

Each farm also chooses how orders are delivered: the farm itself (its own fee, PIN at the door) or Nolifa Delivery.

8. The AI, and what is honest about it

Feature	What it is	Honest limit
Leaf scanner	Trained model on the phone, with an online service when there is signal	A guide, not a lab diagnosis
Cattle symptom checker	Rule-based matcher over 24 symptoms and 14 conditions, weighted toward controlled diseases	Not a vet. Notifiable diseases always say call the state vet
Crop health AI	Agronomic model: crop calendar, adjusted by the farmer's scans and sprinkler watering, with a range and a confidence level	An estimate, not a neural network. It says so, and gets sharper as the farmer scans
Sprinkler planner	Suggests run times against the crop's daily water need	The app cannot switch the hardware on. The farmer runs the schedule
Water forecast	Double exponential smoothing on reservoir readings	Statistical, labelled as such, shows confidence and reason
Community proof	On-phone scan or symptom check plus a photo is required to post	Neighbours confirm with I see this too

9. Water radar

- Provider readings become a 7-day forecast and status: Healthy, Watch, Low, Critical.
- When a community is Low or Critical, people can vote. Each account counts once. When votes reach the community's threshold, the request goes to the provider's desk.
- The provider acknowledges and dispatches trucks with an ETA. Every stage is visible. Municipality updates show in the farmer's community only when a provider has answered.
- Farms with a borehole or tanker can register as suppliers and answer requests.
- Demo mode simulates five communities, neighbours, a provider and a dry spell so the whole loop can be shown on one phone.

10. How it is built

Area	Choice
Framework	Flutter, Riverpod for state, a dark glass design system shared by all apps
Shared package	ubuntu_core: market, water, theme, farms, shop, business and notification interfaces
Storage	On the phone by default (so two phones do not see each other yet). A Firestore version of every repository is built and tested but switched off
Firebase	Off. Switched on with --dart-define=USE_FIREBASE=true by whoever connects the backend

Area	Choice
Maps	flutter_map with free OpenStreetMap tiles. Hexagons are drawn in the app, so no Google Maps key is needed
Notifications	flutter_local_notifications. Welcome back on opening in Grow and Drop
Driver actions	url_launcher for navigation, image_picker for proof photos, injected through providers
Images	Farm photos and logos generated with Higgsfield and bundled as assets

11. Tests and status

Project	Tests	Covers
ubuntu_core	137	Fees, escrow, PINs, disputes, inspection rules, farms and baskets, water forecast and voting, shop, business and driver screens
Nolifa Grow	95	Four tabs, orders, self-delivery, crop advisor and sprinkler plans, community, hexagon map, alarm, water provider, notifications
Nolifa Drop	36	Sign-in, onboarding, three tabs, menu, settings, water pop-up, welcome notification
Nolifa Delivery	25	Sign-in, driver welcome cards, jobs, earnings, account
Firestore rules	100+	Real emulator cases plus contract tests replaying what the apps write

All analyzers report no errors or warnings. All apps were run on the iOS Simulator.

12. What is real and what is simulated

Real in the prototype	Simulated or not built yet
The apps and their offline storage; fee, escrow and PIN rules; disputes logic; inspection rules; crop advisor; hexagon map; community proof rules; notifications on the phone; security rules and their tests	FNB payments and identity checks; vet, surveyor and mediator decisions; neighbour and driver activity in demo mode; water provider readings; the shared backend between phones; delivery proof photo upload; push notifications

13. Weaknesses reviewed, and what changed

Weakness found	What changed
Scope too broad for one pitch	One proposition: Protect, Prepare, Provide. New docs/PITCH.md. Login and welcome text, Grow Home and a Household readiness card in Drop all say it
Water radar depends on others	Water page has Where this comes from: who measures, how fresh, who responds, what if nobody does. A 3-minute demo of one full water disruption, plus the provider's desk. Farms can act as suppliers
Affordability vs free knowledge	A Free for everyone card (scanning, water, alerts, community, buying) beside Paid: selling. Prices marked as hypotheses to test. docs/VALIDATION_PLAN.md
Offline claims unclear	Offline guide: works with no signal, needs a connection, needs another person. Docs and this PDF reworded
Business model needs validation	Simulated parts labelled in checkout, wallet, verification, inspection and water, and listed on Trust and Safety. Validation plan with pilot and tests

14. What the backend team takes over

Feature	Backend job
Shared data between apps	Turn Firebase on, deploy rules, replace the Delivery app's copied Firebase files with its own registration (com.nolifa.delivery)
Inspection results	An inspection service writes passed or failed. A farm must not be able to set its own result
Community posts	A posts collection plus Storage for photos, one confirmation per person
Water suppliers	Approve farms as suppliers, and allow approved suppliers to dispatch
Delivery proof	Upload the photo to Storage and store its URL on the order with a matching rule
Notifications	Push (FCM) for orders, water and outbreaks, honouring Settings toggles
Payments and identity	FNB payments, escrow holding and identity checks in a pilot district

More detail: docs/FIREBASE_HANDOFF.md, docs/INSPECTION.md, docs/LEAN_CANVAS.md, docs/SSDLC.md and docs/JUDGE_QA.md.